

Department Overview Department Name Department of Electrical Engineering Faculty Faculty of Information & Communication Technology (FICT) Establishment Founded in 2012. The graduate (MS) program was subsequently initiated in Spring 2014. Accreditation The undergraduate program is accredited by the Pakistan Engineering Council (PEC) under Level-II, indicating compliance with high-quality education standards. Core Focus Focuses heavily on Electric Power Systems Engineering and Telecommunication Engineering. The curriculum balances mathematical depth and engineering fundamentals with professional skills like personality grooming, public communication, entrepreneurship, and management. Vision To become a global leader in the field of Electrical Engineering having a pivotal role in the development and prosperity of society. Mission To impart quality learning in the field of electrical engineering, promote excellence in research & innovation, and play a leading role in economic, social, and cultural development. Program Educational Objectives (PEOs) The department's educational goals are designed to produce professional engineers who will demonstrate professional competence in electrical engineering through in-depth knowledge and skills, alongside contemporary interdisciplinary studies, engage in continual professional development through effective communication, teamwork, and lifelong learning, and depict professional integrity, practice, and implement ethics and social responsibility to the global community. Bachelor of Science (BS) Program Admission Requirements F.Sc (Pre-Engineering) completed from any recognized board or equivalent with at least 60% marks and Diploma of Associate Engineer (DAE) completed in a relevant engineering field with at least 60% marks. Degree Requirements Total Credit Hours 136 with a semester credit load of 15-18 credit hours per semester and a Minimum Academic Standing of CGPA greater than or equal to 2.0. Specialization Streams Students can choose to specialize in either Telecommunication Engineering or Power Systems Engineering. Curriculum Plan (First 4 Semesters Schema) Note that credit hour listings follow the format (Theory + Lab). 1st Semester includes HUM-163 Functional English in Knowledge Area Humanities with Credit Hours 3+0, MATHP-118 Calculus and Analytical Geometry in Knowledge Area Natural Sciences with Credit Hours 3+0, EE-112/L Linear Circuit Analysis & Lab in Knowledge Area Engineering Foundation with Credit Hours 3+1, CS-123 Applications of ICT in Knowledge Area Computer Sciences with Credit Hours 2+1, ENGG-217L Engineering Drawing in Knowledge Area Engineering Foundation with Credit Hours 0+1, PHY-205/L Applied Physics & Lab in Knowledge Area Natural Sciences with Credit Hours 2+1, and MTH-XX Math Deficiency - I (if applicable) in Knowledge Area Natural Sciences with Credit Hours 3+0 for a Semester Total Credit Hours of 17. 2nd Semester includes MATHA-214 Differential Equations in Knowledge Area Natural Sciences with Credit Hours 3+0, XX Natural Science Elective in Knowledge Area Natural Sciences with Credit Hours 3+0, ELE-102L Electrical Workshop Practice in Knowledge Area Engineering Foundation with Credit Hours 0+1, HUM-XX Pakistan Studies in Knowledge Area Humanities with Credit Hours 2+0, CS-112/L Computer Programming & Lab in Knowledge Area Advanced Computer & IT with Credit Hours 3+1, EE-208/L Electronic Devices & Circuits & Lab in Knowledge Area Engineering Foundation with Credit Hours 3+1, and MTH-XX Math Deficiency II (if applicable) in Knowledge Area Natural Sciences with Credit Hours 3+0 for a Semester Total Credit Hours of 17. 3rd Semester includes MATHA-213 Complex Variables and Transforms in Knowledge Area Natural Sciences with Credit Hours 3+0, HUM-116 Ideology and Constitution of Pakistan in Knowledge Area Humanities with Credit Hours 2+0, EE-110/L Digital Logic Design & Lab in Knowledge Area Engineering Foundation with Credit Hours 3+1, CS214/L Data Structures and Algorithm & Lab in Knowledge Area Advanced Computer & IT

with Credit Hours 3+1, ELE-201/L Electrical Network Analysis & Lab in Knowledge Area Engineering Foundation with Credit Hours 3+1, and ENVIRON-103 Occupational Health and Safety in Knowledge Area Multidisciplinary Eng. with Credit Hours 1+0 for a Semester Total Credit Hours of 18. 4th Semester includes HUM-XX Social Sciences Elective in Knowledge Area Humanities with Credit Hours 2+0, MATHP-111 Linear Algebra in Knowledge Area Natural Sciences with Credit Hours 3+0, HUM-101 Islamic Studies/Ethics in Knowledge Area Humanities with Credit Hours 2+0, MATHA-217 Probability Methods in Engineering in Knowledge Area Engineering Foundation with Credit Hours 3+0, EE-261/L Signals and Systems & Lab in Knowledge Area Engineering Foundation with Credit Hours 3+1, and EE-111 Electromagnetic Field Theory in Knowledge Area Engineering Foundation with Credit Hours 3+0 for a Semester Total Credit Hours of 17. Core Higher Semester Subjects (Semesters 5 to 8 Highlights) for the 5th Semester include Electrical Machines (3+1), Microprocessors and Interfacing (3+1), Communication Systems (3+1), Civics and Community Engagement (2+0), and MDEE Elective (3+0) for a total of 17 Credits. 6th Semester includes Expository Writing (3+0), Linear Control Systems (3+1), Power Distribution and Utilization (3+1), Depth Elective (Core)-1 (3+1), and Project Management (2+0) for a total of 17 Credits. 7th Semester includes Depth Elective-II Core (3+1), Depth Elective-III (3+1), Flexible Elective I (Climate Change and Risk Management, 2+1), Depth Elective-IV (3+0), Arts & Humanities Elective (Muslim Heritage, 2+0), and Final Year Design Project Part-I (0+2) for a total of 18 Credits. 8th Semester includes Entrepreneurship (2+0), Depth Elective-V (3+0), Flexible Elective-II (3+0), Flexible Elective-III (3+0), and Final Year Design Project Part-II (0+4) for a total of 15 Credits. Master of Science (MS) Program Admission Requirements include an Academic Background of 16 years of education or equivalent (such as a 4-year B.E or BS degree) in a relevant field from an HEC-recognized institution with Minimum Grades of at least 60% marks under the annual system or a minimum CGPA of 2.5 out of 4.0 and Testing of a valid GAT General score with at least 50% marks. MS Degree Requirements include a Total Credit Hours of 30 consisting of Coursework of 24 credit hours of graduate-level courses (8 courses) and a Research Thesis of 6 credit hours of research conducted under the supervision of PhD faculty members. Minimum Passing Standing is a CGPA greater than or equal to 2.5. Areas of Research include specialized streams available in Telecommunication Systems Engineering and Power Systems Engineering. Future Program Plans state the department is actively working toward launching a Ph.D. Engineering Program which will broaden specialized research tracks across several fields including Power Systems Engineering, Telecommunication Systems Engineering, Electronic Engineering, Control Systems Engineering, and Computer Systems Engineering.